From Frozen Affine Pairs to Growing CRT Fibers

Uniformity Envelopes, Cutoff Phase Diagrams, and the Fixed-Two Quantifier Barrier

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Abstract

TPC-137 proves full-Möbius logarithmic cancellation for every fixed determinant-two affine pair and fixed periodic coefficient. The actual TPC family is not fixed. This paper measures the gap by comparing two squarefree decompositions. The magnitude cutoff has tail $Y^{o(1)}(N/R+\sqrt{Y})$, combined parameter modulus at most MR_{eff}^4, and point census $O(N+R_{\mathrm{eff}}^2)$. The prime-square cutoff has tail $O(N/P+\sqrt{Y})$, at most $4^{\pi(P)}$ ordered component pairs, and combined parameter modulus

$$M\,\mathrm{lcm}(k,\ell)^2 \le M(P^{\#})^2.$$

For every fixed integer data height Q, Tao's fixed-affine theorem is uniform over the corresponding finite record set. A slow diagonal gives some non-effective $Q_{*,\omega}(x)\to\infty$, but no prescribed growth rate. There is now also an explicit positive corridor: Tao–Teräväinen prove a power-of-log estimate, outside a small logarithmic-density exceptional set, uniformly for positive affine coefficients and constants of sufficiently small polylogarithmic height. This corrects the older "fixed only" frontier, but does not contain an arbitrary actual CRT family and does not control deterministic exceptional prefixes. No ordinary all-prefix power estimate or positive L2 input is claimed.

1 Two cutoff languages

Let I be an integer interval of length N, and let

$$D(z)=d+sz, \qquad V(z)=u+az, \qquad su-ad=2$$

be primitive and positive on I. Suppose

$$\max_{z\in I}\{D(z),V(z)\}\le Y.$$

Let M denote a fixed periodic modulus already present in the coefficient. The word "combined modulus" below refers to the z-parameter. Passing to the ambient n-frame may multiply it by the separately recorded base modulus as.

1.1 Magnitude cutoff

TPC-128 defines

$$Q_R(m)=\sum_{\substack{k\le R\\ k^2\mid m}}\mu(k)$$

and proves the two-mask tail

$$\ll Y^{o(1)}\left(\frac{N}{R}+\sqrt{Y}\right), \tag{1}$$

as well as the point census $O(N + R^2)$ [3]. Only

$$R_{\text{eff}} = \min\{R, \sqrt{Y}\}$$

can occur. After also resolving the period,

$$L_z \le M R_{\text{eff}}^4. \tag{2}$$

The exact-cutoff regime $R \ge \sqrt{Y}$ has zero truncation tail, not merely the bound (1).

1.2 Prime-square cutoff

TPC-137 instead uses

$$\mathsf{S}_P(m) = \prod_{p \le P} (1 - \mathbf{1}_{p^2|m}) = \sum_{k|P\#} \mu(k) \mathbf{1}_{k^2|m}$$

[5]. Its Boolean nature yields

$$\sum_{z \in I} \left| \mu^2(D(z))\mu^2(V(z)) - \mathsf{S}_P(D(z))\mathsf{S}_P(V(z)) \right| \ll \frac{N}{P} + \sqrt{Y}. \tag{3}$$

Proposition 1.1 (Prime-cutoff component geometry)**.** *Before compatibility removes empty components, the prime cutoff has at most*

$$4^{\pi(P)} \tag{4}$$

ordered pairs (k, ℓ). Every nonempty pair, after resolving the period M, has parameter modulus

$$L_z = \operatorname{lcm}(M, k^2, \ell^2) \le M \operatorname{lcm}(k, \ell)^2 \le M(P^\#)^2. \tag{5}$$

Proof. The primorial has $2^{\pi(P)}$ divisors, independently in each coordinate. The identity $\operatorname{lcm}(k^2, \ell^2) = \operatorname{lcm}(k, \ell)^2$ proves (5). □

Remark 1.2 (The exponent correction). The final upper bound in (5) contains $(P^\#)^2$, not $(P^\#)^4$. The latter would count the two divisors as if their prime supports could be duplicated inside the same squarefree primorial.

2 A fixed-height uniform envelope

The fixed theorem of Tao cannot be called on an unspecified growing record [1, 4]. It does, however, give uniformity over every finite bounded record set.

Definition 2.1 (Finite affine record set)**.** For $Q \ge 1$, let $\mathfrak{D}(Q)$ be the finite set of integer records

$$\mathfrak{d} = (a_1, b_1, a_2, b_2, M, r)$$

such that

$$1 \le a_i, M \le Q, \quad |b_i| \le Q, \quad 0 \le r < M,$$

the reduced forms are primitive and positive after a fixed translation, and

$$0 < |a_1 b_2 - a_2 b_1| \le Q.$$

The residue $r \bmod M$ represents one periodic component.

For an admissible terminal ratio $\omega = \omega(x)$, define

$$\mathcal{E}_{\mathfrak{d}}(x,\omega) = \frac{1}{\log \omega} \left| \sum_{x/\omega < z \le x} \frac{\lambda(a_1 z + b_1)\lambda(a_2 z + b_2)\mathbf{1}_{z \equiv r \pmod{M}}}{z} \right|.$$

Theorem 2.2 (Fixed-height uniformity)**.** *For every fixed Q,*

$$\max_{\mathfrak{d} \in \mathfrak{D}(Q)} \mathcal{E}_{\mathfrak{d}}(x, \omega(x)) \longrightarrow 0 \tag{6}$$

as $x \to \infty$, whenever $1 \le \omega(x) \le x$ and $\omega(x) \to \infty$.

Proof. Resolve the one fixed residue in each record. After factoring fixed contents, the two forms remain nonparallel. Tao's theorem gives convergence for every individual record. The maximum in (6) is over the finite set $\mathfrak{D}(Q)$. □

Remark 2.3 (What finiteness does not give)*.* The convergence threshold in theorem 2.2 may depend on Q. Taking $Q = Q(x)$ inside the theorem is invalid unless one has first proved a uniform rate controlling that dependence.

3 A proved small-polylog almost-scale corridor

The post-2025 source frontier is stronger than the fixed theorem. The following is the affine special case recorded after Tao–Teräväinen's quantitative correlation theorem [2, Theorem 3.1 and Remarks 3.2].

Theorem 3.1 (Tao–Teräväinen affine corridor)**.** *There is a sufficiently small absolute $c_0 > 0$ and an exceptional set of scales $\mathcal{E}$ such that*

$$\frac{1}{\log X} \int_{[1,X] \cap \mathcal{E}} \frac{dt}{t} \ll (\log X)^{-c_0}, \tag{7}$$

and, for every $N \notin \mathcal{E}$,

$$\sup_{\substack{1 \le a_i, b_i \le (\log N)^{c_0} \\ a_1 b_2 \ne a_2 b_1}} \frac{1}{N} \left| \sum_{n \le N} \lambda(a_1 n + b_1)\lambda(a_2 n + b_2) \right| \ll (\log N)^{-c_0}. \tag{8}$$

The constant c_0 may be decreased so that one exponent controls both the data height and the saving.

Remark 3.2 (Source quantifier lock)*.* The underlying $\mathcal{L}$-parameter theorem is local in a separate ambient variable X: it produces $\mathcal{E}_X \subset [\sqrt{X}, X]$, controls $N \in [\sqrt{X}, X] \setminus \mathcal{E}_X$, and is uniform only for

$$W \le \mathcal{L}^c, \qquad b, h_1, h_2 = O(\mathcal{L}^c).$$

Its non-pretentious branch assumes, in the notation of the source,

$$\exp M(g_1; X^2, \log^{1/125} X) \gg \mathcal{L}.$$

The source verifies this condition for λ and then records the global affine special case stated in theorem 3.1, using $W = a_1 a_2$ and shrinking the absolute exponent. We invoke that recorded special case, not an extension to arbitrary periodic weights or squarefree masks.

Remark 3.3 (Exact eligibility test). Equation (8) is callable only after every residue split and content extraction has produced positive integer coefficients and constants within its small-polylog height, with nonzero reduced determinant. A dyadic block uses two prefix endpoints; its exceptional set must therefore include the corresponding endpoint dilates. Resolving many CRT moduli also requires the union of their pulled-back exceptional sets to retain vanishing logarithmic density. None of these checks follows merely from saying that the original determinant is two.

Corollary 3.4 (Eligible finite-family transport)**.** *Suppose a scale-N decomposition has total absolute coefficient mass at most $(\log N)^{\kappa_{\text{cen}}+o(1)}$, every reduced record satisfies remark 3.3, and the union of the required pulled-back exceptional sets has logarithmic density $o(1)$. If $\kappa_{\text{cen}} < c_0$, then the retained Liouville part has a power-of-log saving at every remaining scale.*

Proof. Apply (8) to every eligible component and sum absolute values. The resulting exponent is $c_0 - \kappa_{\text{cen}} + o(1) > 0$. ☐

This corollary is a source-safe L1 corridor, not an actual TPC certificate: the literal data-height, exceptional-set union, squarefree tail, weights, phases and prefixes still have to be verified.

4 A qualitative diagonal beyond the explicit corridor

Lemma 4.1 (Non-effective affine diagonal)**.** *For each fixed admissible terminal-ratio function ω, there exists a nondecreasing integer function $Q_{*,\omega}(x) \to \infty$ such that*

$$\max_{\mathfrak{d}\in\mathfrak{D}(Q_{*,\omega}(x))} \mathcal{E}_{\mathfrak{d}}(x, \omega(x)) \longrightarrow 0. \tag{9}$$

No prescribed lower bound on $Q_{*,\omega}(x)$ follows, and the construction need not be uniform in ω.

Proof. For each integer j, use theorem 2.2 to choose X_j so large that the maximum over $\mathfrak{D}(j)$ is at most $1/j$ for $x \geq X_j$. Increase the X_j if necessary and set $Q_{*,\omega}(x) = j$ on $[X_j, X_{j+1})$. The thresholds come from qualitative convergence for the chosen ω, so this construction emits neither an effective formula nor a named growth class. ☐

Corollary 4.2 (Exact containment interface)**.** *Suppose every retained record in an actual family at scale x has encoded height at most $Q_{*,\omega}(x)$, and suppose its complete absolute reassembly cost is $x^{o(1)}$ with a rate compatible with (9). Then the fixed-height conclusions can be summed on that family. Neither premise is supplied by lemma 4.1.*

The compatibility clause is necessary: an unspecified $x^{o(1)}$ factor can grow faster than the unspecified decay in (9).

5 Cutoff phase diagram

The two decompositions serve different goals.

Cutoff	Elementary tail	Parameter modulus	Main obstruction
Magnitude R	$Y^{o(1)}(N/R + \sqrt{Y})$	MR_{eff}^4	divisor maximum, weighted census, growing affine data
Prime P	$N/P + \sqrt{Y}$	$M(P^{\#})^2$	$4^{\pi(P)}$ components and growing data

For example, $P \asymp \log\log x$ gives only a logarithmic tail. More precisely, its $1/P$ relative term then tends to zero only like $1/\log\log x$; the separate square-root term must also be controlled. On the positive side, the crude bounds

$$4^{\pi(P)} \leq 4^P, \qquad P^{\#} \leq P^P$$

show that its precompatibility costs are $x^{o(1)}$. Sharper prime-number estimates show that for a sufficiently small constant multiple of $\log\log x$, the component count is $(\log x)^{o(1)}$ and the primorial modulus is a small power of $\log x$. This creates a genuine candidate for theorem 3.1, conditional on the literal base height and all exceptional-set/reassembly checks.

Conversely, forcing the $1/P$ relative part of the elementary prime tail to be $O(x^{-\delta})$ requires $P \geq x^{\delta}$. Then the cutoff and its record family lie far outside the small-polylog corridor. The effective value cap $k, \ell \leq \sqrt{Y}$ can reduce the realized family, but it does not by itself return it to that corridor.

Definition 5.1 (Growing determinant-two affine gate)**.** A callable growing theorem must specify a function $\varepsilon_{\mathrm{aff}}(x; Q)$ and prove, in its own statement,

$$\sup_{\mathfrak{d} \in \mathfrak{D}(Q(x))} \mathcal{E}_{\mathfrak{d}}(x, \omega(x)) \leq \varepsilon_{\mathrm{aff}}(x; Q(x))$$

for the declared $Q(x)$, origins and windows. An actual H3 return additionally needs the physical weights, phases and prefix quantifiers, or a separately proved transfer for them.

6 Stops and claim boundary

FIXED-DATA STOP. If any coefficient, residue modulus, period or cutoff grows outside a proved envelope, the fixed theorem is not callable.

DIAGONAL STOP. The existence of $Q_{*,\omega}(x)$ does not show that a prescribed actual height is at most $Q_{*,\omega}(x)$. Renaming the slow diagonal as the physical family is forbidden. The diagonal is not the same object as the explicit almost-scale corridor in theorem 3.1.

POWER STOP. The qualitative $o(1)$ in theorem 2.2 has zero x-power exponent. It cannot be entered as $1/400$, nor can it pay a positive physical loss.

Statement	Level and status
Two cutoff identities, tails, moduli and component counts	Exact/elementary L0.
Fixed-height uniform envelope	Unconditional frozen L1.
Small-polylog affine envelope outside a small exceptional scale set	Unconditional sourced L1; actual eligibility open.
Existence of some slow $Q_{*,\omega}(x)$	Exact non-effective diagonal.
Larger prescribed polylogarithmic or power affine envelope	Open beyond the sourced corridor.
Containment of the actual CRT family	Not proved.
Ordinary all-prefix power saving or positive L2	Open; no L2 claim.

TPC-139 therefore locates the next analytic task precisely: prove the growing gate of definition 5.1, or replace the termwise CRT route. Even an almost-scale theorem would still need a selector theorem before it could control deterministic physical prefixes, which is the subject of the next paper.

References

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